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Project Proposal Report

Energy-Robustness Trade-offs in Compressed TinyML Vision Models

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1. Abstract

Model compression is essential for deploying convolutional neural networks (CNNs) on energy- and memory-constrained TinyML and edge devices, but most existing studies report savings using static compression ratio (model file size) rather than measured runtime energy consumption, leaving open the question of whether ratio-based robustness conclusions hold under real energy budgets. Building on a 2025 benchmark study that evaluated quantization, pruning, and clustering against natural image corruptions using compression ratio [1], this project focuses on robustness. A ResNet-18 CNN is trained on CIFAR-10 and compressed using INT8 post-training quantization, unstructured magnitude pruning, and structured channel pruning, each swept across a range of configurations. For each technique, we will try to match configurations that are matched to target energy-reduction levels ($\approx 30\%$ and $\approx 50\%$) and evaluated for clean accuracy and robustness to common corruptions using CIFAR-10-C and the Mean Corruption Error (mCE) metric. Robustness degradation across techniques is compared statistically at matched energy budgets. The expected outcome is an empirical energy-robustness Pareto frontier that offers practical, energy-grounded guidance for compressing vision models for resource-constrained edge deployment, together with evidence on whether compression ratio based robustness rankings generalize to real measured energy savings

2. Introduction and Background

Information and Communications Technology (ICT) currently accounts for 2 to 3% of global CO2 emissions, a figure projected to reach 7 to 8% by 2030. AI workloads are a primary driver of this growth, particularly from running repeated model inference on billions of edge devices like phones, wearables, and embedded sensors (collectively known as Edge AI or TinyML). Because these devices operate under strict battery and memory limits, model compression techniques like quantization and pruning are a mandatory requirement for deployment.

Also, compressed models are deployed into uncontrolled real-world environments where inputs are not clean benchmark data. Visual corruption like fog, motion blur, sensor noise, and compression artifacts are unavoidable in applications like outdoor cameras and autonomous navigation. If a compression technique saves energy but silently degrades a

model's reliability under these conditions, it creates a hidden safety risk that developers might miss if they only evaluate clean test accuracy.

This project investigates how three common compression techniques (INT8 post-training quantization, unstructured magnitude pruning, and structured channel pruning) balance measured energy savings against robustness to natural image corruption in a TinyML-scale convolutional neural network. It directly extends a recent 2025 benchmark study by replacing static compression ratio (file size reduction) with directly measured runtime energy consumption as the primary efficiency metric.

3. Problem Statement

Let f denote a baseline CNN classifier (ResNet-18) trained on the CIFAR-10 training set, with clean test accuracy $\text{Acc}(f)$. Let $C = \{c_1, c_2, c_3\}$ denote the three compression operators under study, INT8 post-training quantization, unstructured magnitude pruning, and structured channel pruning, each applied at intensity level α (e.g., sparsity ratio), producing a compressed model $f(c_i, \alpha)$.

For every compressed configuration, three quantities are measured directly rather than estimated:

1. Energy $E(c_i, \alpha)$ mean measured energy in Joules,
2. Accuracy $\text{Acc}(c_i, \alpha)$ top-1 classification accuracy on the clean CIFAR-10 test set.
3. Robustness $\text{mCE}(c_i, \alpha) = (1/15) \sum \text{CE}_c$, the Mean Corruption Error across the 15 corruption types in CIFAR-10-C, computed relative to the uncompressed baseline's error, following the corruption-error formulation introduced by Hendrycks and Dietterich and adopted directly by the base paper for this project.

Two derived quantities follow,

The energy-reduction ratio

$$\Delta E(c_i, \alpha) = 1 - E(c_i, \alpha) / E(f)$$

and The accuracy drop,

$$\Delta \text{Acc}(c_i, \alpha) = \text{Acc}(f) - \text{Acc}(c_i, \alpha)$$

The research problem is, at targeted energy-reduction levels eg. $\{30\%, 50\%\}$, the effect on robustness of the compression techniques.

4. Motivation

1. The most relevant state-of-the-art study [1] evaluates multiple compression techniques against corruption robustness but relies on a static file compression ratio rather than instrumented energy consumption and lacks support for newer architectures.
2. Recent benchmarks examine only one compression family in isolation, including a quantization-only robustness benchmark [2] and a pruning-only calibration study [3], without comparing techniques against each other or measuring energy.
3. A 2025 study performs rigorous hardware-based energy measurement via a high-precision power meter [4], but it focuses exclusively on large language models and standard NLP benchmarks without addressing corruption robustness or CNN vision applications.
4. No major existing work jointly measures real hardware-level inference energy, corruption robustness, and accuracy across multiple compression families for vision models, which is the precise empirical gap this project addresses.

5. Literature Review

5.1 Base Paper

Da Silva, Pereira, Barboza, dos Santos Neto, and Ribeiro (2025), “Evaluating the Impact of Compression Techniques on the Robustness of CNNs under Natural Corruptions.” [1]

This paper is adopted as the base paper because it is the most recent and most directly comparable prior work: it applies quantization, pruning, and weight clustering individually and in 16 combinations to ResNet-50, VGG-19, and MobileNetV2, evaluates them on CIFAR-10-C and CIFAR-100-C using the Mean Corruption Error (mCE) metric, and reports a Pareto-optimal multi-objective analysis across accuracy, mCE, and compression ratio. Its central finding is that compression often preserves or even improves robustness relative to the uncompressed baseline, but that the effect is technique and architecture dependent. Critically for this project, the paper's efficiency axis is compression ratio (file size), it uses a TensorFlow/LiteRT pipeline, and its own stated future work recommends exploring PyTorch and deeper hyperparameter sweeps both of which this project adopts directly. This project reuses the base paper's CIFAR-10-C corruption set, its mCE formulation, and its training protocol (Adam optimizer, categorical cross-entropy, early stopping) to keep results comparable, while substituting measured energy for compression ratio as the efficiency axis.

5.2 Supporting References

Xiao, Liu, Zhang, Qin, Guo, and Liu (2023), “RobustMQ: Benchmarking Robustness of Quantized Models,” Visual Intelligence. [2]

Benchmarks quantized models against adversarial attacks, natural corruptions, and systematic noise, finding that lower-bit quantization improves adversarial robustness but reduces robustness to natural corruption and systematic noise motivating this project's focus on natural corruption specifically (rather than adversarial robustness) as the metric most relevant to real-world edge deployment conditions, and confirming that quantization's robustness effects are not uniformly positive.

Mitra, Schwalbe, and Klein (2024), “Investigating Calibration and Corruption Robustness of Post-hoc Pruned Perception CNNs,” CVPR Workshops. [3]

Shows that post-hoc pruning of CNN image classifiers can improve calibration and corruption resilience, evaluated across structured and unstructured pruning at sparsity rates from 0% to 70%. This project's pruning sparsity swep (30%, 50%, 70%) is modelled on this protocol, and its scope is extended here by adding quantization as a directly compared alternative and by adding a measured-energy axis that this paper does not include.

Husom, Goknil, Astekin, Shar, Kåsen, Sen, Mithassel, and Soylu (2025), “Sustainable LLM Inference for Edge AI: Evaluating Quantized LLMs for Energy Efficiency, Output Accuracy, and Inference Latency,” ACM Transactions on Internet of Things. [4]

Provides the methodological basis for energy measurement, combining hardware power monitoring and software profiling to evaluate energy, latency, and accuracy trade-off. This project adapts the same dual measurement approach for CNN-based vision inference and additionally incorporates robustness-to-corruption evaluation.

Hendrycks and Dietterich (2019), “Benchmarking Neural Network Robustness to Common Corruptions and Perturbations,” ICLR. [5]

Introduces the CIFAR-10-C/ImageNet-C benchmarks and the Corruption Error (CE) and mean Corruption Error (mCE) metrics used to quantify robustness, serving as the foundational reference for robustness evaluation.

5.3 Synthesis of the Gap

Study	Techniques Compared	Robustness Type	Measured Energy?	TinyML-scale Focus?
Da Silva et al. [1] (Base Paper)	Quant. + Pruning + Clustering	Natural corruption	No (ratio only)	Partial (MobileNetV2)
Xiao et al. [2]	Quantization only	Natural + adversarial	No	No
Mitra et al. [3]	Pruning only	Natural corruption + calibration	No	No
Husom et al. [4]	Quantization (LLMs)	Not studied	Yes (hardware)	Edge, not TinyML-CNN
This Project	Quant. + 2 Pruning types	Natural corruption	Yes (hardware-measured)	Yes (TinyML-scale CNN)

6. Research Objectives

1. Measure energy consumption (Joules) of a baseline TinyML-scale CNN (ResNet-18) and its quantized and pruned variants on GPU hardware.
2. Evaluate clean-test accuracy and robustness (mCE on CIFAR-10-C) for each compression configuration.
3. Identify, for each compression technique, the configuration that achieves matched target energy-reduction levels ($\approx 30\%$ and $\approx 50\%$) within an acceptable accuracy-loss bound ($\leq 3\%$).
4. Statistically compare robustness degradation across the three compression technique at matched energy budget
5. Construct an empirical energy-robustness Pareto frontier to provide practical deployment guidance for TinyML/edge vision systems, and compare it against the compression-ratio-based frontier reported in the base paper.

7. Research Questions

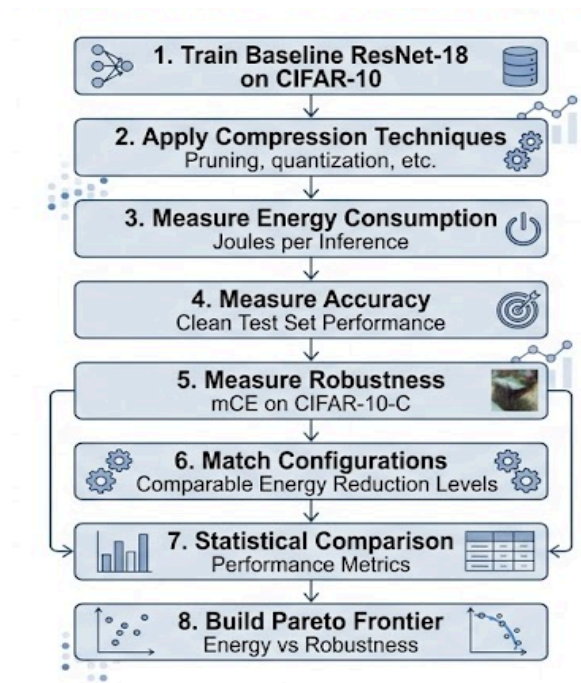
7.1 Primary Research Question

For TinyML-scale CNNs, how do different compression techniques, such as quantization, magnitude pruning, and structured pruning, that achieve comparable energy reductions differ in their impact on robustness to natural image corruption?

7.2 Sub-Research Questions

1. What is the measured per-inference energy consumption (Joules) of ResNet-18 under each compression technique and configuration, relative to the uncompressed baseline?
2. At matched energy-reduction levels of $\approx 30\%$ and $\approx 50\%$, does mean Corruption Error (mCE) on CIFAR-10-C differ significantly between INT8 quantization, magnitude pruning, and structured pruning?
3. Which compression technique yields the most favorable Pareto-optimal balance among measured energy reduction, clean-accuracy retention, and corruption robustness?

8. Methodology



8.1 Dataset

CIFAR-10 (50,000 training / 10,000 test images, 10 classes) is used for training and clean-accuracy evaluation. CIFAR-10-C, the same corruption benchmark used by the base paper, is used for robustness evaluation; it applies 15 corruption types (e.g., Gaussian noise, motion blur, fog, JPEG compression) at 5 severity levels to the CIFAR-10 test images. To keep the experiment tractable within the project timeline, a representative subset of 5 corruption types spanning the noise, blur, weather, and digital

categories is evaluated at severities 3 and 5, with the full 15-type sweep included as a stretch goal if time permits.

8.2 Baseline Model and Training Protocol

ResNet-18 (CIFAR-adapted stem) is trained using PyTorch, following the recommendation in the base paper's future-work section. Training uses the Adam optimizer, categorical cross-entropy loss, and early stopping (patience = 5, maximum 30 epochs). The baseline model's clean accuracy is checked against the base paper's reported results before compression.

8.3 Compression Techniques

1. **T1:** INT8 Post-Training Quantization: implemented via PyTorch's native quantization API.
2. **T2:** Unstructured Magnitude Pruning: implemented via `torch.nn.utils.prune`, swept at sparsity levels {30%, 50%, 70%}.
3. **T3:** Structured (Channel) Pruning: filter/channel pruning by magnitude, swept at the same sparsity levels {30%, 50%, 70%} for direct comparison with T2.

8.4 Energy Measurement Protocol

Energy is measured on Kaggle GPU hardware using GPU power telemetry (`pynvml/nvidia-smi`) and the CodeCarbon profiler. After discarding 10 warm-up runs, $N = 30$ single-image inference passes are timed and power-sampled. Energy per inference (Joules) is computed by integrating power $\times$ time. Mean energy, 95% confidence intervals, and the energy-reduction ratio (ΔE) relative to the uncompressed baseline are reported.

8.5 Robustness Evaluation Protocol

For each compressed configuration, accuracy is evaluated on the selected CIFAR-10-C corruption subset, and Mean Corruption Error (mCE) is computed following the base paper's formulation, relative to the uncompressed baseline's error rate on the same corruptions.

8.6 Matching Energy Budgets Across Techniques

For each technique, sparsity sweeps (plus the INT8 quantized model) are used to identify configurations closest to the target energy-reduction bands ($\sim 30\%$ and $\sim 50\%$).

Interpolation between adjacent sparsity levels is applied when needed. This produces matched-energy model sets for direct cross-technique comparison.

8.7 Statistical Analysis Plan

1. Shapiro-Wilk and Levene's tests assess normality and variance homogeneity.
2. If assumptions are met, one-way ANOVA with Tukey HSD post-hoc testing is used; otherwise, Kruskal-Wallis with Dunn's post-hoc testing is applied.

8.8 Synthesis and Visualization

Results are synthesized into an energy–robustness Pareto frontier (energy reduction on one axis, mCE on the other, clean accuracy encoded by color), directly modeled on the base paper's own multi-objective figure but with measured energy substituted for compression ratio. Comparing this frontier against the base paper's ratio-based frontier provides a direct, visual test of whether the two efficiency metrics lead to the same practical recommendations, the central question motivating this project

8.9 Tools

PyTorch (training and compression), pynvml / NVIDIA power telemetry and CodeCarbon (energy measurement), SciPy / pingouin / statsmodels (statistical analysis and power calculation), and matplotlib / seaborn (visualization) all consistent with the toolset recommended in the course materials.

9. Expected Results and Outcomes

1. Energy-robustness Pareto frontier for compression techniques.
2. Best compression method for TinyML and edge deployment.
3. Determine whether compression ratio is a reliable proxy for real energy, or if direct energy measurement is required.

10. References

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